EE115 Analog Circuits
模拟电路

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Outline

- PN junction
- SEDTRA/SMITH book page 148-167；

PN Junction

- P-type semiconductor in contact with n-type
- Basic building blocks of semiconductor devices
  - Diodes
  - Bipolar junction transistors (BJT)
  - Metal-oxide-semiconductor field effect transistors (MOSFET)

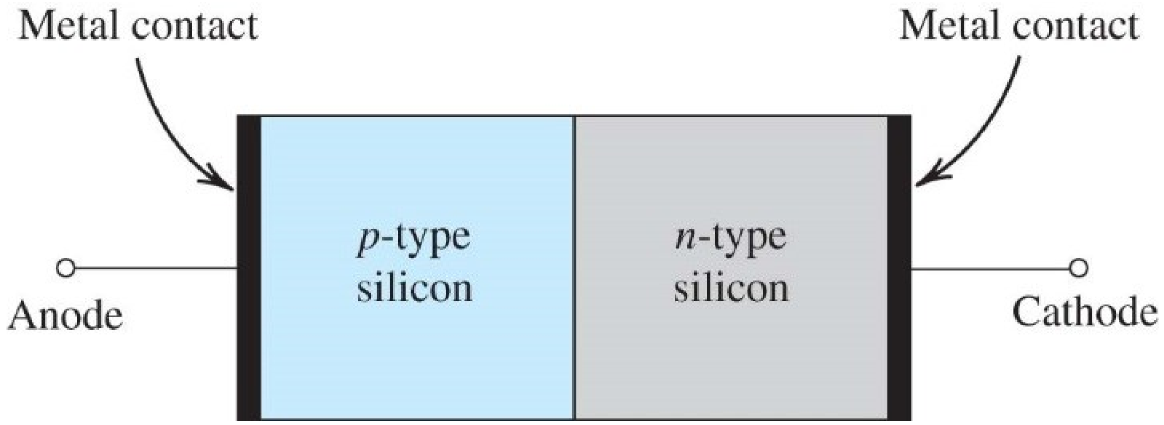


Figure 3.8 Simplified physical structure of the *pn* junction. (Actual geometries are given in Appendix A.) As the *pn* junction implements the junction diode, its terminals are labeled anode and cathode.

Built-in Voltage

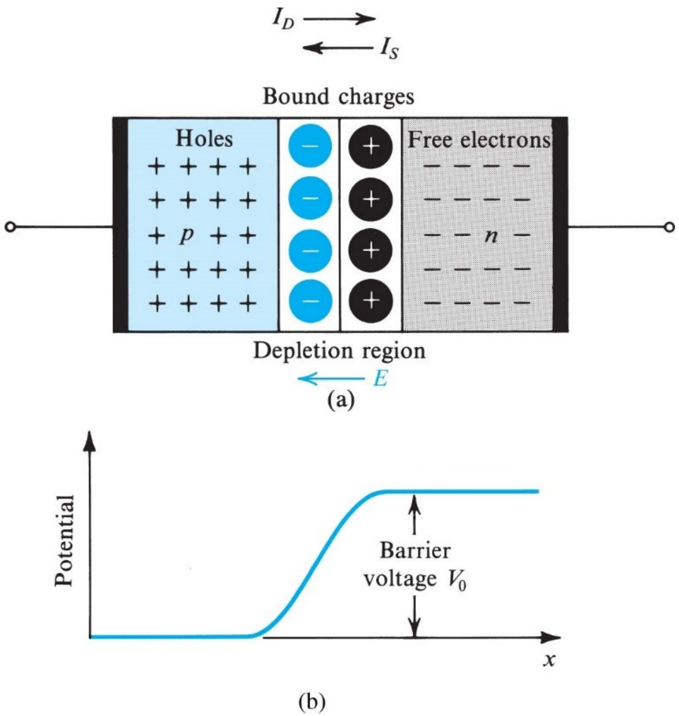


Figure 3.9 (a) The *pn* junction with no applied voltage (open-circuited terminals). **(b)** The potential distribution along an axis perpendicular to the junction.

Example:

- At pn junction, free electrons from n-side "recombine" with free holes from p-side.
- "Depletion region (耗尽区)" has no electrons or holes, but has fixed (immobile) charges from donor and acceptor ions
- The fixed charges establish an electric field, and create a potential difference between p and n-sides.

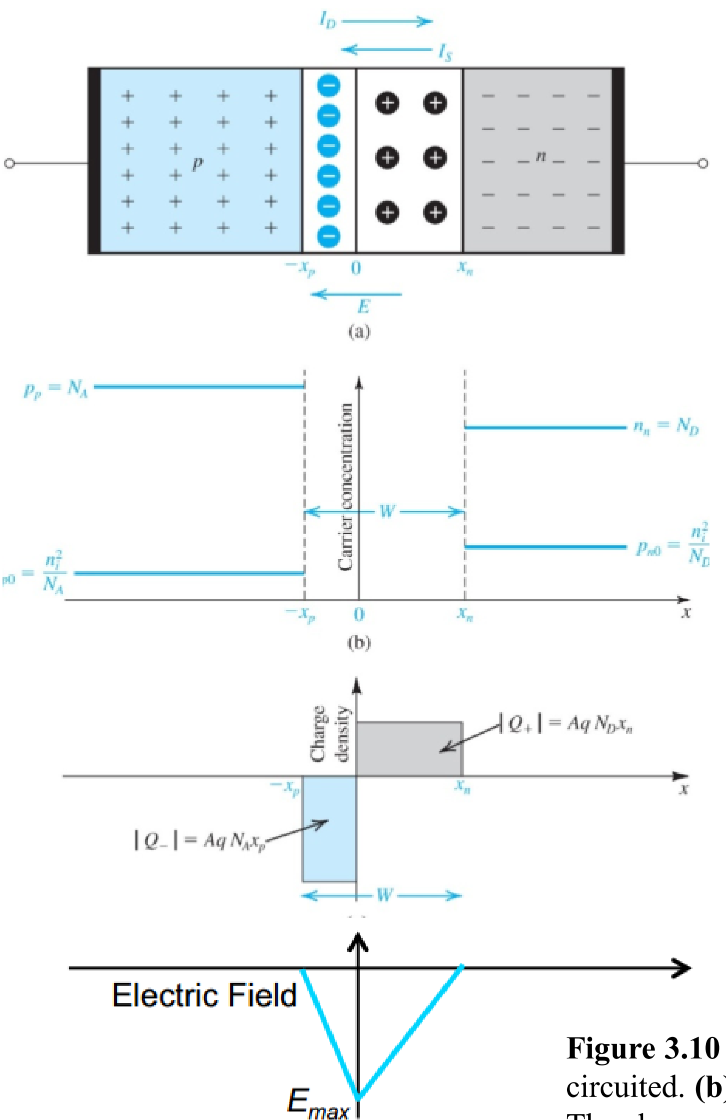
$$V_o = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

- This potential is called "built-in potential(内建电势)"
  - V_T: thermal voltage (= 26 mV at room temp)
  - N_A : acceptor concentration on p-side
  - N_D : donor concentration on n-side
  - n_i : intrinsic carrier concentration (=1.5×10^{10} cm^{-3} at room temp)

Electrostatic Analysis of PN Junction

$$E(x)=\begin{cases}\dfrac{-qN_A(x+x_p)}{\varepsilon_S}, & -x_p<x<0\\[2mm] \dfrac{qN_D(x-x_n)}{\varepsilon_S}, & 0<x<x_n\end{cases}$$

$$V(x)=-\int_{-x_p}^{x}E(x')dx'$$



- *Charge in the depletion regions:*

$$|Q_+| = qAx_nN_D \qquad |Q_-| = qAx_pN_A$$

- *Equaling the charge:*

$$qAx_nN_D = qAx_pN_A \qquad \frac{x_n}{x_p}=\frac{N_A}{N_D}$$

- *Width of depletion region:*

$$W = x_n + x_p = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0}$$

- *Then we have:*

$$x_n = W\frac{N_A}{N_A+N_D}$$

$$x_p = W\frac{N_D}{N_A+N_D}$$

Charge stored in the depletion region

$$Q_J = |Q_+| = |Q_-|$$

$$Q_J = Aq\left(\frac{N_AN_D}{N_A+N_D}\right)W$$

$$Q_J = A\sqrt{2\epsilon_s q\left(\frac{N_AN_D}{N_A+N_D}\right)V_0}$$

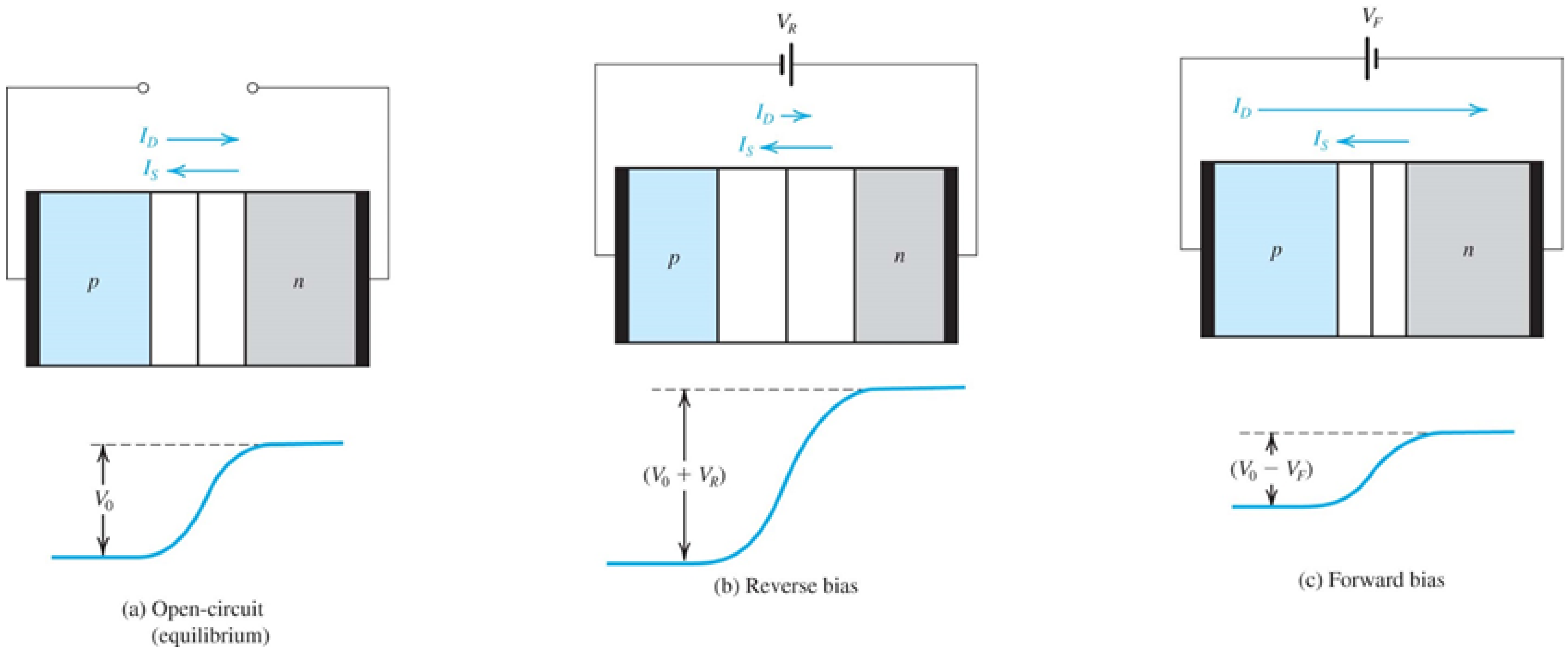
Figure 3.10 (a) A *pn* junction with the terminals open-circuited. **(b)** Carrier concentrations; note that $N_A > N_D$. **(c)** The charge stored in both sides of the depletion region; $Q_J = |Q_+| = |Q_-|$. **(d)** The electric field.

Depletion Width Under Bias

$$W = \sqrt{\frac{2\varepsilon_S}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_o - V)}$$

V is the applied voltage to the pn junction, it is positive for forward bias and negative for reverse bias.

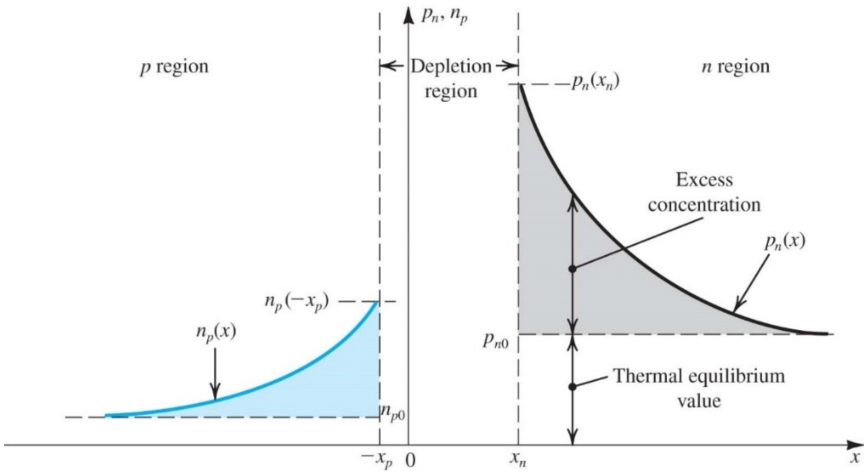
Depletion width is widened in reverse bias



(a) equilibrium; **(b)** reverse bias; **(c)** forward bias

Figure 3.11 The *pn* junction in:

Current-Voltage (I-V) Characteristics



$$J_n(-x_p) = q\left(\frac{D_n}{L_n}\right)n_{p0}\left(e^{V/V_T} - 1\right)$$

Figure 3.12 Minority-carrier distribution in a forward-biased pn junction. It is assumed that the p region is more heavily doped than the n region; $N_A >> N_D$

- Under forward bias, minority carriers at the edge of the depletion region is boosted up by diffusion current
- *Hole concentration at n region*

$$p_n(x) = p_{n0} + p_{n0}\left(e^{V/V_T} - 1\right)e^{-(x-x_n)/L_p}$$

- *Current density due to hole diffusion:*

$$J_p(x) = -qD_p\frac{dp_n(x)}{dx} = q\left(\frac{D_p}{L_p}\right)p_{n0}\left(e^{V/V_T} - 1\right)e^{-(x-x_n)/L_p}$$

$$= q\left(\frac{D_p}{L_p}\right)p_{n0}\left(e^{V/V_T} - 1\right)$$

- *Total diffusion current:*

$$I = A\left(J_p + J_n\right)$$

$$I = Aq\left(\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right)\left(e^{V/V_T} - 1\right)$$

- *Finally, we get the I-V equation:*

Substituting for $p_{n0} = n_i^2/N_D$ and for $n_{p0} = n_i^2/N_A$ gives

$$I = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)\left(e^{V/V_T} - 1\right)$$

- *Saturation current:* $I_S = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$

I-V Curve

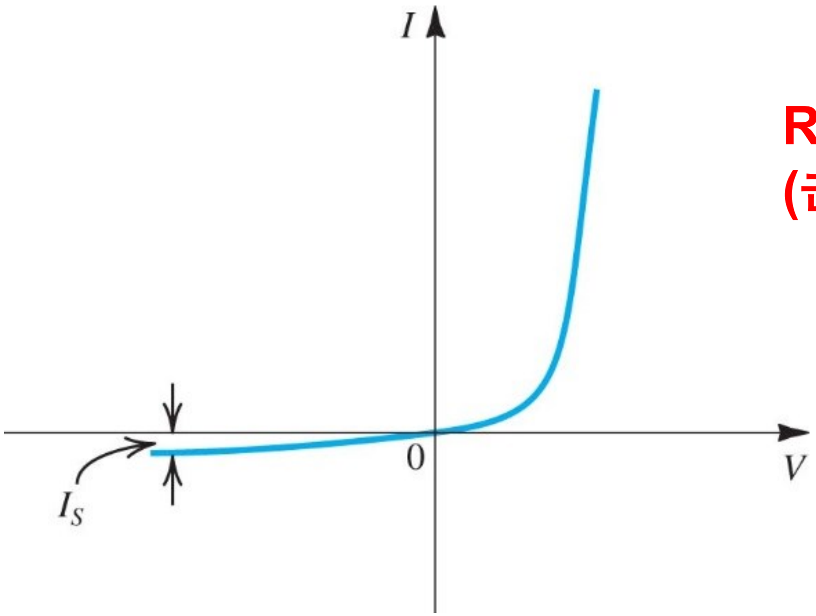


Figure 3.13 The *pn* junction *I–V* characteristic.

$$I = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)(e^{V/V_T} - 1)$$

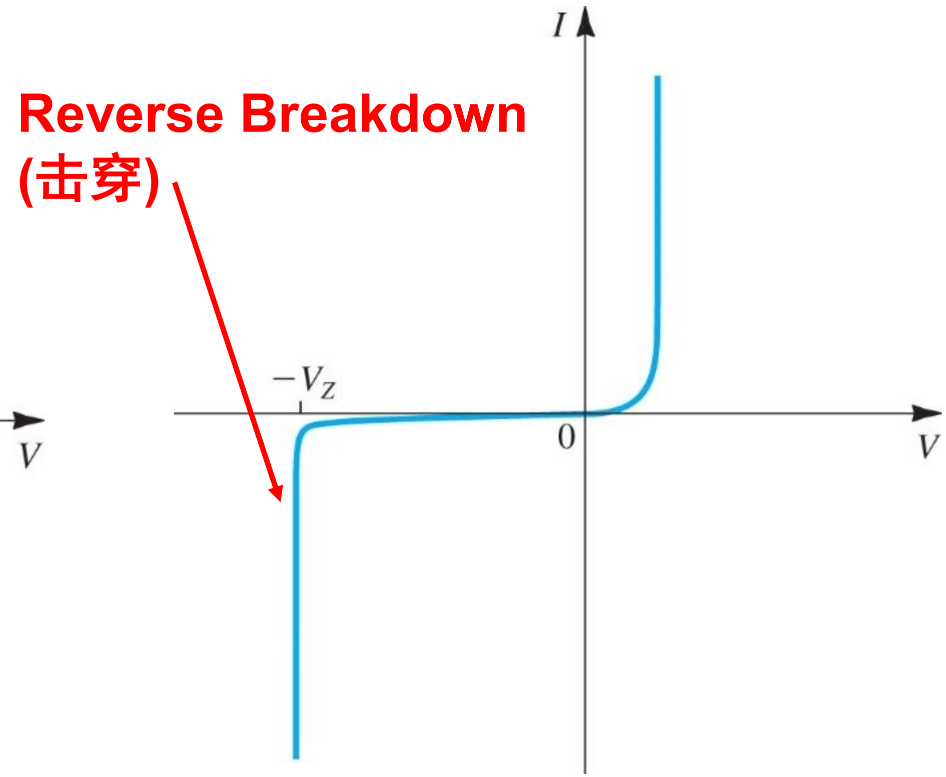


Figure 3.14 The *I–V* characteristic of the *pn* junction showing the rapid increase in reverse current in the breakdown region.

Depletion Capacitance (mainly reverse bias)

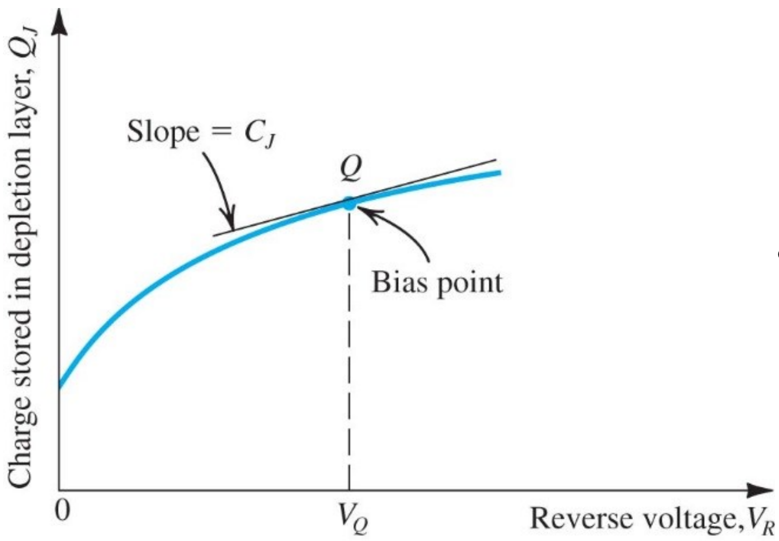


Figure 1.42 The charge stored on either side of the depletion layer as a function of the reverse voltage V_R.

- *Charge in either depletion region:*

$$Q_J = A\sqrt{2\epsilon_s q \frac{N_A N_D}{N_A + N_D}(V_0 + V_R)}$$

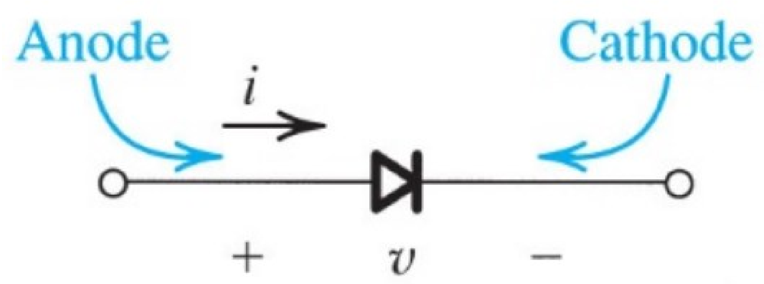
- *Capacitance at specific reverse-bias:*

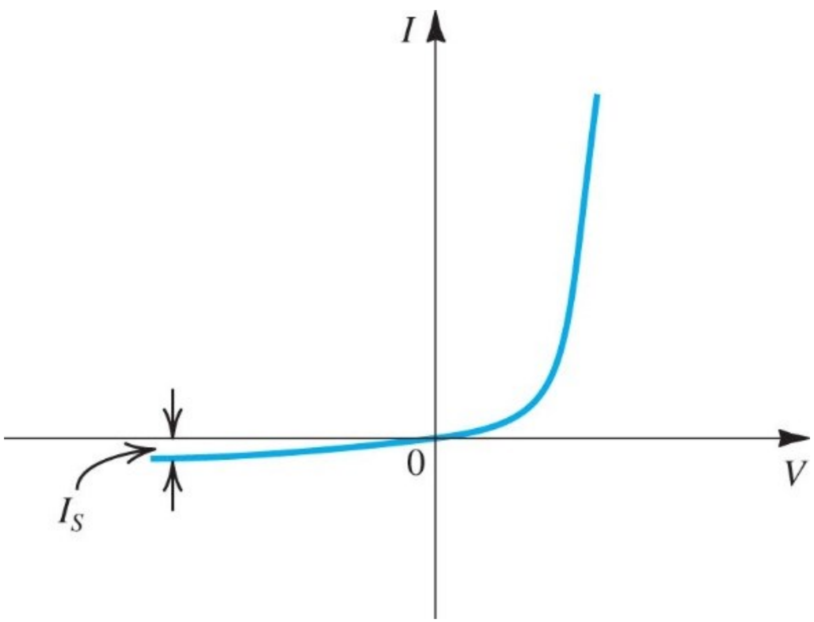
$$C_j = \left.\frac{dQ_J}{dV_R}\right|_{V_R = V_Q} = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}}$$

$$C_{j0} = A\sqrt{\left(\frac{\epsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}$$

Variable capacitance controlled by reverse voltage

Summary of PN Junction





Built-in potential

<u>Forward bias</u>
- I-V curve
- Diffusion capacitance

<u>Reverse Bias</u>
- Negligible current
- Depletion capacitance

Other important parameter
- Depletion Width

Appendix: Derivation of pn Junction Potential

$$E(x)=\begin{cases}\dfrac{-qN_A(x+x_p)}{\varepsilon_S}, & -x_p<x<0\\[2ex] \dfrac{qN_D(x-x_n)}{\varepsilon_S}, & 0<x<x_n\end{cases}$$

$$V(x)=-\int_{-x_p}^{x}E(x')dx'$$

<u>(1) for $-x_p<x<0$</u>: $V(x)=-\int_{-x_p}^{x}E(x')dx'=\int_{-x_p}^{x}\dfrac{qN_A(x'+x_p)}{\varepsilon_S}dx'=\dfrac{qN_A}{2\varepsilon_S}\left(x'+x_p\right)^2\Big|_{x'=-x_p}^{x'=x}=\dfrac{qN_A}{2\varepsilon_S}\left(x+x_p\right)^2$

<u>(2) for $0<x<x_n$</u>: Because $E(x)$ has different expression for $x<0$ and $x>0$, the integration should be performed in two separate ranges, first from $-x_p$ to 0, and then from 0 to x. We can use $V(x=0)$ from the above equation for the first intgration. Therefore,

$$V(x)=\frac{qN_A}{2\varepsilon_S}x_p^2-\int_0^x\frac{qN_D(x'-x_n)}{\varepsilon_S}dx'=\frac{qN_A}{2\varepsilon_S}x_p^2-\frac{qN_D(x'-x_n)^2}{2\varepsilon_S}\Bigg|_0^x$$

$$=\frac{qN_A}{2\varepsilon_S}x_p^2-\left(\frac{qN_D(x-x_n)^2}{2\varepsilon_S}-\frac{qN_Dx_n^2}{2\varepsilon_S}\right)=\frac{qN_A}{2\varepsilon_S}x_p^2+\frac{qN_Dx_n^2}{2\varepsilon_S}-\frac{qN_D(x-x_n)^2}{2\varepsilon_S}$$

Built-in potential: $V_0=V(x_n)=\dfrac{q}{2\varepsilon_S}\left(N_Ax_p^2+N_Dx_n^2\right)$